

CSE / EEE / ETE 499A (Section 01)

**Project Ethical and Professional Responsibility
in Project Development (CO6)**

**Project Title: Bangla Voice Activated Customer Relationship
Management (CRM) System**

Submitted To

Dr. Shazzad Hosain (SZZ)

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Group No: 03

Group Members

ID	Name
1921647042	Md. Rayhan Chowdhury
2121365642	Fiad Sarowar
2131302642	Tahsina Mamun Niha
2232404642	KAZI AYMAN RAFID SACHCHA

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Ethical and Professional Responsibility in Project Development

Abstract

This report fulfills the CO6 requirement for CSE/EEE/ETE 499A by documenting the ethical and professional responsibilities undertaken throughout the development of the Bangla Voice Activated Customer Relationship Management (CRM) System. The project is a mobile-first, Bengali-language voice CRM built with Flutter and FastAPI that records Bangla speech, transcribes it using a Hugging Face BanglaASR model, extracts structured CRM fields through rule-based NLP, and stores leads locally in SQLite for full CRM operations including status tracking and CSV export.

The scope of this report covers every phase of the software lifecycle—problem definition, system design, implementation, testing, deployment, and maintenance—through the lens of ethical engineering principles drawn from the IEEE Code of Ethics and the ACM Code of Ethics and Professional Conduct. The report analyses stakeholder obligations, legal compliance within Bangladesh's evolving digital regulatory framework, a structured risk assessment with mitigation strategies, and professional conduct expectations for the team. A candid reflection on lessons learned is also included. The central argument of this report is that responsible engineering is not a final-stage checklist but a continuous discipline that must be embedded in every technical decision from requirements gathering to production maintenance.

1. Introduction

1.1 Project Overview

The Bangla Voice CRM Management system is a Bengali-language, voice-driven mobile application designed to democratize customer relationship management for small and medium enterprises (SMEs) in Bangladesh. Field sales agents and customer service representatives interact with the system entirely in spoken Bangla: they record or upload audio, the backend ASR service transcribes the speech, and a Dart-based NLP module extracts structured CRM fields—customer name, phone number, address, location, lead type, product interest, and support issue. Leads are stored locally in a SQLite database and can be exported to CSV for sharing and analysis. The dashboard provides real-time statistics on total leads, sales vs. support breakdown, and average extraction confidence.

1.2 Why Ethical and Professional Responsibility Matters for This Project

A voice-based CRM operating in the field handles a particularly sensitive class of data: it captures spontaneous human speech, which may inadvertently include information beyond what the user intends to share; it stores names, phone numbers, and addresses; and it makes automated consequential decisions—

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such as classifying a lead as a sales lead or a support ticket—that directly affect business operations and customer experience. These characteristics make ethical responsibility not merely a compliance exercise but a practical engineering necessity.

Three intersecting factors elevate the ethical stakes further. First, the system is intended for deployment in a context where data-subject literacy about digital rights is still developing, making consent and transparency even more important than in mature markets. Second, the legal framework governing personal data, cybersecurity, and electronic evidence in Bangladesh is actively evolving, and a system designed today must be capable of adapting to regulations that will crystallise over the next few years. Third, the combination of voice recording and automated NLP-based decision-making creates capabilities that can be misused if the system is not designed with principled constraints from the outset.

1.3 Report Structure

This report is organized as follows. Section 2 enumerates the ethical principles and professional standards that guide the analysis. Section 3 provides a detailed stakeholder analysis including potential ethical conflicts. Section 4 examines ethical considerations across each phase of the software development lifecycle. Section 5 addresses legal and regulatory compliance specific to Bangladesh. Section 6 presents a structured risk assessment and mitigation plan. Section 7 discusses professional responsibility of the team. Section 8 provides a reflective account of lessons learned. Section 9 concludes with recommendations. References are listed in Section 10.

2. Ethical Principles and Professional Standards

2.1 Core Ethical Principles

The following ethical principles have been selected as directly relevant to the Bangla Voice CRM project. Each principle is stated and then contextualised for this specific system.

2.1.1 Privacy and Data Minimisation

Privacy is the right of individuals to control information about themselves. In this system, every interaction generates potentially sensitive data: the voice recording itself, the transcript (which may contain more contextual information than a typed form), and the extracted CRM fields (customer name, phone number, address). The principle of data minimisation requires that the system collects, stores, and retains only the data strictly necessary for the stated CRM purpose.

Concretely, this means: audio files should not be retained indefinitely after transcription; transcripts should not be shared beyond the intended CRM use; and no data should be used for secondary purposes (e.g., behavioural profiling or employee surveillance) without explicit consent.

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2.1.2 Informed Consent and Transparency

Users—both the agents operating the app and the customers whose information is captured—must be clearly informed about what data is being collected, for what purpose, how long it will be retained, and who will have access. For the agents, this means explicit permission prompts before the app accesses the microphone or location services. For customers whose voice data or addresses are recorded, the agent has an ethical obligation to inform them at the point of data capture. Transparency also means that automated decisions—such as the system classifying a lead as a support ticket rather than a sales lead—are visible to the user and can be corrected.

2.1.3 Safety and Non-Maleficence

The system must not cause harm. In the CRM context, this includes the risk of incorrect NLP extraction leading to a wrong phone number being stored and an unintended person being contacted, or an incorrectly classified lead type causing a customer's support complaint to be treated as a sales opportunity. Audio validation, confidence scoring, and manual override mechanisms form the core of the safety architecture. More broadly, the system must not become a tool for unfair monitoring of agents or discrimination based on inferred characteristics from voice data.

2.1.4 Fairness and Inclusivity

The system must function equitably across the diversity of its intended user base. This includes: speakers with different regional accents of Bangla (Sylheti-influenced, Chittagonian-influenced, Noakhali-influenced); users in noisy urban environments versus quieter settings; older smartphone models with lower microphone quality; and users with limited literacy who rely entirely on voice input. Systematic underperformance for any of these groups would constitute an unfair outcome. The team has a duty to test with varied speakers and report accuracy differentials honestly rather than citing only best-case performance.

2.1.5 Accountability and Integrity

The team is accountable for the system's behaviour. This extends to acknowledging limitations in ASR accuracy, being honest about the boundaries of prototype versus production readiness, documenting known failure modes, and providing a clear correction path when the system makes mistakes. Integrity requires that evaluation results not be cherry-picked, that training and test data not be contaminated, and that claims made in documentation accurately reflect the system's observed behaviour.

2.1.6 Sustainability and Responsible Resource Use

Third-party APIs—particularly the BanglaASR model via the Hugging Face inference endpoint—carry usage costs, rate limits, and dependency risks. The system should use these resources proportionately, avoid wasteful polling, cache results where appropriate, and have a documented fallback strategy if the API

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becomes unavailable. From an environmental perspective, backend GPU computation for ASR should be scheduled efficiently rather than triggered redundantly.

2.2 Professional Standards Framework

This report is aligned with two internationally recognised professional ethics frameworks.

The IEEE Code of Ethics (2020 revision) commits IEEE members and, by extension, engineering practitioners, to: hold paramount the safety, health, and welfare of the public; treat all persons fairly; avoid injuring others, their property, reputations, or employment; be honest and realistic in stating claims; reject bribery; improve technical competence; seek, accept, and offer criticism; avoid conflicts of interest; protect privacy; and support colleagues in following the code.

The ACM Code of Ethics and Professional Conduct (2018) organises obligations around four pillars: General Ethical Principles (honesty, avoiding harm, privacy, fairness, respecting intellectual property, honouring confidentiality); Professional Responsibilities (high quality work, competence, assessment and acceptance of reviews, non-discrimination, awareness of potential harms); Leadership Principles (for those in leadership roles, which includes the team lead in this project); and Compliance with the Code.

Both frameworks converge on the principle that engineering decisions are never purely technical—they carry ethical weight that practitioners are professionally obligated to consider, disclose, and address.

3. Stakeholder Analysis

3.1 Identification of Stakeholders

A stakeholder is any individual, group, or institution that affects or is affected by the system. For the Bangla Voice CRM, the following stakeholders have been identified and analysed.

Stakeholder	Role / Relationship	Expectations	Primary Ethical Concerns
Field Sales Agents / Customer Service Representatives	Primary end-users who record voice, manage leads, and update lead status	Reliable app, accurate extraction, clear UI in Bangla, fair performance evaluation	Privacy of their own interactions; risk of being penalised by incorrect automated decisions
Customers / Data Subjects	Individuals whose names, phone numbers,	Consent before recording, accurate and limited data	Uninformed data collection; potential misuse of contact data

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Stakeholder	Role / Relationship	Expectations	Primary Ethical Concerns
	and addresses are captured	handling, right to correction	
SME Business Owners / Managers	Deploy the system and use dashboards to monitor leads and sales outcomes	Accurate extraction, reliable reports, data security, cost-effectiveness	Over-reliance on automation; legal liability if system is misused
Project Team	Design, implement, test, and document the system	Academic integrity, professional conduct, honest reporting	Plagiarism; overclaiming accuracy; inadequate testing; unsafe code practices
Supervisor / Institution	Academic oversight and evaluation	Honest work, proper documentation, compliance with course requirements	Misrepresentation of results; ethical violations in project conduct
Future Commercial Deployers	Potential businesses that may adapt this prototype for production	Operational viability, compliance readiness, well-documented architecture	Liability if prototype assumptions do not hold at scale
Society and Regulators	Broader public interest; regulatory bodies such as BTRC	Technology that respects rights, complies with law, and does not enable data misuse	Normalisation of unregulated voice data collection; privacy erosion

3.2 Ethical Conflicts Between Stakeholders

Stakeholder interests are not always aligned. The following conflicts have been identified and must be navigated carefully.

3.2.1 Data Collection Volume vs. Privacy

Business owners may prefer maximum data collection—retaining all audio, transcripts, and location logs indefinitely—to have a complete operational record. Customers, however, have a legitimate interest in data minimisation and limited retention. The system's design must favour the more privacy-protective position by default, with any expanded retention requiring explicit consent.

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3.2.2 Automation Convenience vs. Verifiable Accuracy

Agents may prefer that extracted CRM fields are accepted without review, as manual verification adds a step to their workflow. Managers, however, need assurance that the data saved in the CRM is accurate and has been reviewed by a human. These goals conflict because reducing friction for the agent increases the risk of incorrect data entering the system. The design choice of displaying all extracted fields in editable form before saving is a deliberate trade-off in favour of accuracy and accountability over pure convenience.

3.2.3 Demonstration Optimism vs. Honest Disclosure

Academic project presentations create an incentive to showcase best-case performance. Professional integrity requires that the team communicate real-world accuracy—including failure cases, edge cases, and known limitations—rather than presenting a curated demo as representative performance. This is especially important for ASR accuracy and geofencing reliability, both of which degrade meaningfully in adverse conditions.

3.2.4 Maximum Data Collection vs. Customer Privacy

From a business analytics perspective, a manager might benefit from retaining all audio recordings, full transcripts, and detailed interaction logs to analyse agent performance and customer behaviour over time. However, customers whose information is captured have a legitimate interest in data minimisation—only the structured CRM fields that serve the stated purpose should be retained, and raw audio should be discarded after transcription. The system must favour the more privacy-protective position by default, with any expanded retention requiring explicit, informed consent from all parties.

4. Ethical Considerations Across the Development Lifecycle

4.1 Problem Definition and Requirement Analysis

The project addresses a genuine access problem: Bengali-speaking SME workers are excluded from CRM tooling by language barriers, input method friction, and cost. However, the way requirements are defined can itself introduce or mitigate ethical risks.

Inclusivity in requirements: Initial assumptions about the user base—for example, assuming urban users with modern smartphones, standard Dhaka Bangla pronunciation, and reliable internet connectivity—would systematically disadvantage users in Chittagong, Sylhet, or rural areas, as well as older device users. To mitigate this, the requirements explicitly include manual text input as a fallback, a confidence score that flags low-quality extractions for manual review, and support for uploaded audio rather than only live recording.

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Scope discipline: The project's requirements were carefully bounded to prevent feature creep that could introduce new ethical risks. Proposals that emerged during development—such as voice sentiment analysis to assess customer mood or agent tone—were deferred because they would process additional personal attributes beyond the stated CRM purpose and could disadvantage agents unfairly if their tone was misclassified.

Stakeholder consultation: Requirements for the voice extraction module were informed by observations of how local field agents actually capture customer information. This consultation revealed that agents often speak in mixed Bangla-English (code-switching), which motivated the inclusion of English product names in the extraction dictionary and the provision of a manual correction field before any data is saved.

4.2 System Design Phase

The design phase is where ethical principles must be translated into architectural decisions. Four design principles guided this phase.

4.2.1 Privacy-by-Design

Privacy considerations were built into the architecture rather than added afterwards. Sensitive fields (phone numbers, addresses) are stored only in the local SQLite database on the user's device and are never sent to any third-party server. The ASR backend receives only the audio file and returns only the transcript; it does not store audio or transcripts after processing. All network communication between the Flutter client and the FastAPI backend uses HTTPS in the development environment, and the architecture is documented with the expectation of TLS certificate pinning in production.

4.2.2 Transparent Automation

Every automated decision in the system is surfaced to the user. The extraction confidence score is displayed on the lead creation screen. Individual field values extracted from the transcript are shown in editable text fields, allowing the user to review and correct before saving. The delivery verification status (Pending, In Progress, Verified, Failed) is shown on the lead detail screen with a map preview. No consequential action is taken silently without the user's awareness.

4.2.3 Minimal Permission Footprint

The app requests microphone permission only when the voice recording screen is first opened, not at app launch. Both file picker and storage permissions are requested only when the agent attempts to upload an existing audio file. Both permissions display the system-level permission dialogue, which explains what the app will access. This is consistent with the principle of contextual integrity: permissions are requested in the context where they are needed and only for the duration of that need. The app does not request any permissions that are not directly required for its stated functionality.

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4.2.4 Fail-Safe Defaults

When the ASR service is unavailable or returns a low-confidence transcript, the system defaults to prompting the user for manual input rather than saving potentially incorrect automated data. When extraction confidence falls below 50%, the lead creation screen displays a warning banner prompting the user to review each field carefully before saving. These fail-safe defaults ensure that errors in the automated pipeline result in user review rather than silent incorrect outcomes.

4.3 Implementation Phase

4.3.1 Responsible Use of External Libraries and APIs

The project depends on several open-source libraries (Flutter framework, FastAPI, sqflite, file_picker, share_plus) and a third-party API (BanglaASR via Hugging Face). Professional responsibility requires respecting the licences of all dependencies. Flutter and FastAPI are licensed under permissive open-source licences (BSD and MIT respectively). The Hugging Face model is used via the inference API under the terms of the model card. The team reviewed all relevant licence terms and confirmed that academic, non-commercial use is permitted for all dependencies.

All external dependencies are documented in the project's requirements.txt and pubspec.yaml. No code has been copied from external sources without attribution. Functions adapted from Stack Overflow or community documentation are commented with the source URL.

4.3.2 Secrets Management

The FastAPI backend must be protected against unauthorised access that could expose customer voice data or transcripts. During the prototype phase, the backend is run locally without authentication, which is acceptable for development but would constitute a serious security vulnerability in any networked deployment. For production, the correct architecture requires JWT-based authentication for all API endpoints, rate limiting to prevent abuse, and HTTPS with TLS certificate pinning. These requirements are documented in the deployment readiness checklist in Section 4.5 as explicit pre-conditions for any production release.

4.3.3 Code Quality and Traceability

Code changes were managed using Git with descriptive commit messages. Each major feature (audio upload, delivery tracking, Barikoi integration) was developed on a feature branch and merged after peer review within the team. This practice ensures that the history of the codebase is traceable, that regressions can be identified and reverted, and that responsibility for each change is attributed. The team avoided the practice of committing large, undifferentiated 'big bang' changes that make review and debugging difficult.

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4.4 Testing and Validation Phase

Testing carries a specific ethical weight because test results are the primary evidence used to make claims about system performance. Misrepresenting test results—through data contamination, cherry-picking, or failure to test adversarial cases—is a form of professional dishonesty.

4.4.1 ASR and Extraction Testing

The ASR backend was tested with audio recordings from four speakers with different accents (Dhaka, Sylhet, Chittagong, and Comilla regional varieties) and under three noise conditions (quiet room, street background noise, and market crowd noise). Results show that Word Error Rate (WER) degrades from approximately 13–16% in quiet conditions to 22–31% in high-noise conditions. These figures are reported candidly in the CO5 technical report. The CRM field extraction confidence score was found to correlate well with actual extraction correctness (fields extracted with confidence above 70% were correct in approximately 84% of cases in the validation set).

4.4.2 Edge Case and Adversarial Testing

Beyond standard functional testing, the team conducted targeted testing of edge cases that could result in ethically problematic outcomes. These included: audio recordings that capture partial or ambiguous information (e.g., a customer mentioning two phone numbers); transcripts where the ASR misrecognises a name due to a strong regional accent; lead types where the speech contains both sales interest and a support complaint simultaneously; and phone numbers spoken in Bangla digit words that must be normalised before regex extraction. Each edge case was documented, tested, and assigned a defined system behaviour—typically a low confidence score prompting user review rather than a silent incorrect save.

4.4.3 Ethical Limits of Testing

Testing involved only consenting team members acting as test users. No real customer data was collected during testing. Audio recordings made during development were deleted after testing was complete. Location data collected during geofencing tests was stored only in the local test device and was not uploaded to any server.

4.5 Deployment and Maintenance

Although the project is currently at prototype stage, the team has documented a deployment readiness checklist that addresses the ethical and security requirements for production use.

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Deployment Requirement	Current Status	Required for Production
HTTPS-only API communication	Implemented in development	TLS certificate pinning recommended
Authentication and authorisation	Not implemented (prototype)	JWT-based auth with role-based access control
Data encryption at rest	Not implemented	SQLCipher for SQLite encryption
Audit logging	Basic console logging	Structured audit trail with tamper-evident timestamps
Data retention policy	No policy defined	Configurable retention with auto-deletion of audio and transcripts
Incident response plan	Not documented	Defined escalation and notification procedure
Privacy notice to users	Not implemented	In-app onboarding with explicit consent flow
ASR backend hardening	Local dev server, no auth	Rate limiting, input validation, authenticated endpoints
Regulatory compliance review	Discussed in this report	Legal review against finalised Bangladeshi data protection law

Maintenance obligations extend beyond bug fixing. The team acknowledges that the legal environment in Bangladesh is actively changing (see Section 5) and that compliance requirements may shift after deployment. Monitoring for new regulatory developments, updating the app to accommodate new ASR model versions, and reviewing whether third-party API terms have changed are ongoing professional obligations.

5. Legal and Regulatory Compliance

5.1 Constitutional Framework

Article 43 of the Constitution of the People's Republic of Bangladesh recognises the right of citizens to privacy of correspondence and other means of communication, subject to reasonable restrictions imposed by law in the interests of the security of the State, public order, or morality. While this provision directly concerns state interference rather than private actors, it establishes privacy as a constitutionally recognised

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value that informs the interpretation of other legislation and sets a public policy expectation that private systems should also respect.

5.2 Digital Security and Cybersecurity Legislation

Bangladesh's digital legal landscape has undergone significant change. The Digital Security Act, 2018 was repealed and replaced by the Cyber Security Act, 2023, which was itself replaced by the Cyber Security Ordinance, 2025. The Ordinance addresses a range of digital offences but does not constitute a comprehensive personal data protection regime. Its provisions relevant to this project include prohibitions on unauthorised access to computer systems, unlawful interception of data, and misuse of electronic information.

The specific implications for the Bangla Voice CRM are: (a) the FastAPI backend must not be exposed without authentication, as an unsecured endpoint storing customer voice data could constitute a system vulnerability exploitable under the Ordinance's provisions; (b) any log data that includes personal identifiers must be protected against unauthorised access; and (c) the team must not store or process data in ways that could facilitate identity fraud or harassment.

5.3 Emerging Personal Data Protection Framework

As of the time of writing, Bangladesh does not yet have a comprehensive enacted personal data protection law comparable to the European General Data Protection Regulation (GDPR) or India's Digital Personal Data Protection Act, 2023. However, the Bangladesh government has publicly engaged with draft personal data protection frameworks in 2024–2025, and legislation is anticipated in the near to medium term. Regulatory guidance from the Bangladesh Telecommunication Regulatory Commission (BTRC) is also relevant to the collection of phone numbers and the operation of communications-adjacent services.

Given this transitional environment, the safest and most professionally responsible approach is to design the system to comply with the emerging framework's expected core requirements: lawful basis for data processing, informed consent, data minimisation, purpose limitation, storage limitation, data subject rights (access, correction, deletion), and security obligations. This design-for-compliance posture ensures that the system does not require fundamental re-engineering when legislation is finalised.

5.4 Location Data and Employment Law Considerations

The delivery tracking module raises employment law questions in addition to privacy ones. In many jurisdictions, employer tracking of employee location is subject to specific consent, notification, and proportionality requirements. While Bangladesh does not currently have sector-specific legislation on employee monitoring, the general principle that workers must be informed of and consent to workplace monitoring applies as a matter of fair employment practice. For future commercial deployment, businesses

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using the delivery tracking feature should obtain written consent from delivery personnel, provide a clear policy on how tracking data is used, and ensure that tracking data is not used as the sole basis for disciplinary action without manual verification.

5.5 Intellectual Property and Licence Compliance

The project uses open-source software under various licences. Flutter (BSD 3-Clause), FastAPI (MIT), SQLite (public domain), and the sqflite plugin (MIT) all permit free use, modification, and distribution provided attribution requirements are met. The BanglaASR model on Hugging Face is used under its model card terms, which permit research and academic use. The Barikoi API is used under its developer terms of service. The team has reviewed all licence terms and confirmed that academic, non-commercial use is permitted without restriction for all dependencies. A production commercial deployment would require a formal licence review, particularly for the Barikoi API, which has pricing tiers for commercial applications.

6. Risk Assessment and Mitigation

6.1 Risk Register

The following risk register identifies the primary ethical and technical risks of the system, assesses their severity and likelihood, and specifies mitigation measures and residual risk.

Risk	Severity	Likelihood	Mitigation	Residual Risk
Unauthorised exposure of customer personal data (name, phone, address)	High	Medium	Local-only storage; authentication for API endpoints; encryption at rest in production; minimal API surface.	Low-Medium (prototype); must be resolved before production.
ASR bias against non-Dhaka Bangla accents causing systematic extraction errors	Medium	High	Manual correction interface; confidence score display; testing with diverse speakers; honest accuracy reporting.	Medium – model limitations cannot be fully eliminated at prototype stage.
Silent or inaccurate CRM field extraction leading to	Medium	High	User review screen before saving; editable extraction fields; confidence threshold warning.	Low – user review provides a meaningful catch.

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Risk	Severity	Likelihood	Mitigation	Residual Risk
incorrect lead records				
Audio data retained beyond necessity	Medium	Medium	Backend deletes audio after transcription; no long-term audio storage in current design.	Low – verify in production audit.
Unsecured ASR backend endpoint exposing customer audio or transcripts	High	High (current)	Authentication and rate limiting must be added before any networked deployment. Documented as known production gap.	High (prototype) – must be remediated before any public deployment.
Regulatory non-compliance as Bangladesh data protection law evolves	High	High (over time)	Design aligned with expected legal requirements; compliance review scheduled when legislation is finalised.	Medium – depends on final legislative text.
Dependency on third-party Hugging Face ASR API availability	Medium	Low-Medium	Documented fallback to manual text entry; possibility of self-hosted model for production.	Medium – self-hosting addresses dependency but increases infrastructure cost.
Misuse of exported CSV data (e.g., selling customer contact lists)	High	Low	Organisational policy control; data handling policy to be included in deployment documentation.	Medium – technical enforcement is limited; policy and training required.
Code-switching (Bangla-English) causing extraction failures	Medium	Medium	English product names included in extraction dictionary; manual correction field always available.	Low-Medium – coverage will improve with dictionary expansion.

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6.2 Contingency Planning

In addition to the primary mitigations documented above, the following contingency plans address failure scenarios that have a meaningful probability of occurring in field use.

- **ASR backend unavailability:** The Flutter app falls back to the manual text input workflow. A user-facing banner informs the agent that voice processing is temporarily unavailable and that they should type the lead details. Leads created via fallback are flagged in the database with a source field value of 'manual_fallback'.
- **Low extraction confidence:** If the confidence score is below 50%, the lead creation screen displays a red warning banner and highlights each extracted field in amber, prompting the user to verify every value before saving. The user cannot bypass this review step.
- **ASR hallucination or nonsense transcript:** The backend validates audio length and energy level before running the ASR model. If the audio is too short, silent, or contains only noise, the backend returns an error code and the app prompts the user to re-record rather than attempting extraction on a meaningless transcript.
- **Incorrect extraction causing wrong customer contact:** If a user reports that incorrect data was saved—for example, the wrong phone number was extracted and an unintended person was contacted—the system provides a full edit-and-re-extract workflow from the lead detail screen. The edit history is logged in the database for accountability.

7. Professional Responsibility of the Team

7.1 Team Accountability Structure

Professional responsibility is most effective when it is distributed across a team with clear ownership. The Bangla Voice CRM team structured accountability as follows.

Team Member	Primary Responsibility	Ethical Accountability
Md. Rayhan Chowdhury	Flutter frontend, UI/UX, CRM dashboard, lead management	Accessible design; permission UX; transparent status display; Bangla-language interface correctness
Fiad Sarowar	FastAPI ASR backend, confidence scoring, testing, API integration	Secure endpoint design; Honest accuracy reporting; testing with diverse speakers; fair performance evaluation

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Team Member	Primary Responsibility	Ethical Accountability
KAZI AYMAN RAFID	Bangla NLP extraction , audio processing pipeline, API integration	audio data handling; no unauthorised retention; input validation,used logical extraction
Tahsina Mamun Niha	Database design, CSV export, lead status management, documentation	Data minimisation in schema; correct status transition logic; accurate documentation of limitations

While individual ownership is assigned, ethical responsibility is collective. All team members reviewed each other's code for security concerns, privacy risks, and accuracy claims before submission. Disagreements about design choices were resolved through team discussion with reference to the ethical principles in Section 2.

7.2 Honest Communication of Limitations

One of the most practically challenging aspects of professional responsibility is communicating system limitations to stakeholders who may have an interest in hearing optimistic claims. The team commits to the following standards of honest communication.

- ASR accuracy is reported as a range (13–31% WER depending on conditions) rather than a single best-case figure. Demonstrations use audio recorded in conditions representative of real field use, not studio-quality recordings.
- The extraction module is described as an assistive tool, not an infallible classifier. Its limitations for code-switched speech and unusual proper nouns are explicitly documented.
- The prototype is clearly distinguished from a production-ready system. The deployment readiness checklist in Section 4.5 is shared with any future deployer to make clear what additional engineering work is required before public release.
- Known gaps—including the absence of backend authentication and at-rest encryption in the prototype—are documented rather than hidden.

7.3 Academic Integrity

The project report and all submitted deliverables represent the team's own work. All external sources are cited. Code adapted from open-source examples is attributed in comments. No results have been fabricated or altered. The team has not submitted substantially the same work for credit in another course.

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7.4 Continuous Professional Development

Ethical and professional responsibility does not end with project submission. The team recognises that the practices developed in this project—threat modelling, privacy-by-design, honest testing, stakeholder analysis—are transferable skills that should be applied in future professional work. The experience of navigating Bangladesh's evolving regulatory environment also illustrates that engineers must stay informed about legal and policy developments relevant to their work.

8. Reflection and Lessons Learned

8.1 Voice Input Amplifies Privacy Concerns

The most significant insight from this project is that voice input, while dramatically more accessible and efficient for the target user group, also amplifies privacy risks in ways that are not immediately obvious. A user filling in a CRM form types exactly the fields that are asked for. A user speaking freely into a microphone may include far more contextual information—tone, emotion, background conversation, incidentally mentioned third parties—than they intend to share. This makes the design of the audio-handling pipeline (record locally, process server-side, delete immediately after transcription) not merely a technical preference but an ethical necessity.

8.2 Automated Decisions Require Human Review Gates

One of the clearest lessons from implementing the NLP extraction module is that automated decisions should never be final without a human review opportunity. Early in development, a design was considered where the system would automatically save extracted fields immediately after transcription, to minimise the number of steps in the agent's workflow. This was rejected because it would have meant that an ASR error—a misheard name, a transposed digit in a phone number—would enter the CRM without any opportunity for correction. The final design inserts a mandatory review screen between extraction and saving, which adds one step but dramatically reduces the risk of incorrect data propagating through the system. This experience confirms the principle that the appropriate role of automation is to assist and accelerate human decision-making, not to replace it entirely.

8.3 Confidence Scores Are an Ethical Tool, Not Just a Technical Metric

The confidence score—computed as a function of how many of the four key CRM fields (name, phone, address, product) were successfully extracted—initially appeared to be purely a technical output. During testing, however, the team recognised that it also serves an ethical function: it is the primary mechanism by which the system signals to the user that they should exercise extra care before saving. A low confidence score triggers a visible warning and highlights uncertain fields in the UI. Without this signal, users would

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have no objective indicator of whether the automated extraction could be trusted for a given recording. This reframes the confidence score as a transparency mechanism—a means by which the system is honest with the user about the limits of its own accuracy.

8.4 Prototype Security Gaps Must Be Explicitly Documented

During development, several security shortcuts were taken that are appropriate for a local prototype but would be unacceptable in any networked or production deployment: the FastAPI backend runs without authentication, customer data is not encrypted at rest, and there is no rate limiting on the ASR endpoint. Each of these gaps was identified during the team's internal security review and documented explicitly in the deployment readiness checklist. The lesson is that security gaps in prototypes do not become acceptable through familiarity—they must be named, documented, and assigned a clear remediation path so that future developers or deployers cannot overlook them.

8.5 Regulatory Uncertainty Requires Adaptable Design

The absence of a finalised personal data protection law in Bangladesh creates genuine uncertainty for the project. The team's response—designing for compliance with the expected principles of consent, minimisation, purpose limitation, and security—is the professionally appropriate one, but it also means that the design may need to be revised when legislation is enacted. Building adaptability into the architecture (modular consent flow, configurable retention periods, documented data flows) is both an ethical and a practical engineering virtue.

9. Conclusion

The Bangla Voice CRM Management system addresses a genuine and significant access gap in customer relationship management technology for Bengali-speaking SMEs in Bangladesh. Its ethical and professional responsibility profile is shaped by two sensitive capabilities: voice recording and automated NLP-based decision-making. This report has demonstrated that each carries specific ethical obligations—for privacy, fairness, transparency, accountability, and safety—and has documented how the team has addressed those obligations across the full software development lifecycle.

The project is not a finished product; it is a well-designed, honestly evaluated, and responsibly documented prototype. Its value as an academic deliverable lies not only in demonstrating technical competence but in demonstrating the ethical reasoning and professional discipline that make technology trustworthy. The deployment readiness checklist, the risk register, the stakeholder conflict analysis, and the honest accuracy reporting collectively constitute a model for how responsible engineering should be practised.

CO6: Demonstrate an understanding of ethical and professional responsibility in project development phases.

The team commits to carrying the ethical framework developed in this project into future professional work: to always consider who is affected by a system and how, to design for the most privacy-protective configuration that meets functional requirements, to communicate limitations honestly, and to remain informed about the legal and social context in which technology operates.

10. References

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